

CHE 494

Selected Topics in Chemical Engineering

1. Course number and name

CHE 494 Entrepreneurship in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: One, 165 min/wk. lecture; Engineering.

3. Instructor's or course coordinator's name

Said Al-Hallaj

4. Text book, title, author, publisher, year

Selected publications, reports and handouts.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The course provides an introductory view into the various aspects involved in technology commercialization (i.e. taking an idea from the lab to the marketplace).

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand fundamentals and basic concepts of intellectual property, confidentiality of Non-Disclosure Agreements (NDA), patent applications, copyrights, Know-how
- Learn about UIC resources for entrepreneurship; Office of Technology Management, Invention Center, Business Plan Competition, Illinois Venture
- Learn how to develop a business development strategy, business plan, marketing strategies, finance and accounting
- Understand the fundamentals and basic concepts of starting a business, legal structure (LLC, Inc., operating agreements, governance
- Learn concepts and practical strategies for fundraising, grants, seed/angel investment, venture capital, strategic partnerships
- Present a Case Study with final project term paper and presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	X			
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Anderson Entrepreneurship Quiz
- Protecting your intellectual property (IP) (Guest Lecturer: Mark Krivchenia, UIC-OTM)
- UIC Resources Overview (Guest: Mark Krivchenia, UIC-OTM)
- Business Development Strategy (Mike Pintar, COO NETenergy)
- Why Marketing Study is Important
- Financials Basics Steve Wolff, CFO AllCell Technologies, LLC
- Starting Your Own Business (is it a good idea?)
- How to build an effective team
- Corporate Entrepreneurship
- Pitching your idea
- Fundraising Strategy (are you ready to pitch your idea?)